

Coding & Solution

Date	1 July 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	GitHub Repository - Emotion Detection & Learning Support Engine
Programming Language(s)	Python
Framework(s) Used	Streamlit, PyTorch, Hugging Face Transformers
Key Features Implemented	Emotion classification using BiLSTM and BERT, mixed-emotion breakdown, rule-based keyword enhancement, Gemini AI-generated personalized tips and next steps, dual-model comparison view, CSV interaction logging, emotion trend analytics dashboard, AI response toggle.
Pending / Incomplete Features	User authentication, database integration, multilingual support, cloud deployment.
Setup / Run Instructions	1. Clone repository. 2. Create virtual environment. 3. Install dependencies using pip <code>install -r requirements.txt</code> . 4. Run the app using <code>streamlit run app.py</code> .

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project follows a modular architecture with separate data preprocessing, model training, inference, and Streamlit UI layers. Both BiLSTM and BERT emotion classifiers were evaluated side by side, and Gemini AI was integrated for generating personalized guidance. The application was tested across all five emotion categories and ran without critical errors. The project is fully functional and ready for further enhancements such as authentication and cloud deployment.